Table A1: Mesh resolution, computational cost, and number of simulations used in the UQ workflow. SA denotes sensitivity analysis. CPU cost is reported per CFD simulation.

Case	Mesh	# cells	Wall units	CPU h/run	# runs
$4 \times$ Channel flow	Coarse	9.6×10^4	$y^+_{\text{avg}} \lesssim 1$	$\sim$2	4×1408 (SA) +222 (surrogate)
Annular flow	Coarse	1.4×10^4	$r^+_{\text{avg}} \approx 1.2$	$\sim$11	1408 (SA)
	Medium	5.8×10^4	$r^+_{\text{avg}} \approx 0.85$	$\sim$62	222 (surrogate)
Cantilevered rod	Coarse	1.9×10^6	$r^+_{\text{inner,avg}} \approx 0.85$ $z^+_{\text{tip,avg}} \approx 24$	$\sim$2000	65 (surrogate)